

Amit Kiran Rege

PhD Candidate, Computer Science · CU Boulder · Advisor: Prof. Claire Monteleoni

amit.rege@colorado.edu · github.com/amitrege · amitrege.github.io · +1 (303) 960-4240

Research in theoretical machine learning, with a focus on AI safety and control: applying RL theory to LLM post-training, foundational multi-agent bandits, and the mathematics of interpretability. Published at AISTATS, L4DC (oral), IEEE CDC, and Physical Review Applied; 6 papers currently under submission.

RESEARCH

Reinforcement Learning Theory · Multi-Agent Bandits · Data Attribution for Interactive Learning · AI Interpretability & Safety · Generative Models

SELECTED PUBLICATIONS

- **Multi-Agent Lipschitz Bandits** AISTATS 2026
- **Flickering Multi-Armed Bandits** L4DC 2026 **[Oral]**
- **A Unified Framework for Locality in Scalable MARL** L4DC 2026
- **Incentivized Lipschitz Bandits** IEEE CDC 2025
- **Where Did Your Model Learn That? — Label-free Influence for SSL** arXiv 2024
- **Hamiltonian Learning using Machine Learning** Physical Review Applied 2024
- + 6 papers under submission: RL attribution theory, oracle separations for RL, interpretability auditability, KL-regularized agent monitoring

OPEN SOURCE PROJECTS

tinygrad-classroom	Interactive autograd: scalar neuron → backprop, built for teaching	amitrege/tinygrad-classroom
NeuroSteer	Stepwise activation steering library for Llama-style decoder LLMs	amitrege/NeuroSteer
tracemap	Training-data influence with query heatmaps for image classifiers	amitrege/tracemap
rl-attr-code	Exact RL data attribution distinguishing replay vs. interventional effects	amitrege/rl-attr-code

EXPERIENCE

- Investment Associate — Deming Center Venture Fund, CU Boulder** Apr 2025 – Present
 - Source and evaluate early-stage deep-tech companies for CU Boulder’s student venture fund
 - Led investment in Great Sky (AI chip startup); closed at fund’s highest valuation ever (\$50M)
- Student Engineer — DARPA Subterranean Challenge, Team MARBLE** Summer 2019
 - Built object detection & localization pipeline for robots navigating lightless underground mines
 - Placed **4th of 11 teams** worldwide against NASA JPL, CMU, and UPenn
- Research Intern — École Normale Supérieure (advised by Prof. Cezara Dragoi)** Feb – Jul 2018
 - Formally verified round-based synchronicity of Raft and Viewstamp consensus protocols

EDUCATION

PhD, Computer Science · University of Colorado Boulder	2020–2026 (expected)
MS, Computer Science · University of Colorado Boulder	GPA: 4.0/4.0 · 2020
BE (Hons.), Computer Science · BITS Pilani–Goa	2018

SKILLS & TOOLS

Python · PyTorch · JAX · NumPy · C++ | RL Theory · Bandits · AI Interpretability · Generative Models · Formal Verification · LLM Fine-tuning

TALKS & INDUSTRY ENGAGEMENT

- LLMs from Scratch** — Spencer Fane LLP (Industry Talk) Nov 2025
- Frontiers of Generative AI** — Ensemble Innovation Ventures Jan 2026
- Theoretical Machine Learning** — KOA 94.1 FM iHeart Radio Jan 2025
- Guest Lecturer:** CSCI 7000 (Adv. ML), CSCI 4622 (ML), CSCI 5802 (Data Science)